

2. Build Dataset DataFrame

```
Loading image paths...
Found 70,000 Real images
Found 9,001 Fake images

Dataset Summary
label
0      15000
1       9001

Total Images : 24,001

Extra Dataset detected, merging...

=====
Final Dataset
=====
Train      : 51,199
Validation : 6,400
Test       : 6,401

=====
Stratification / Class-Balance Check
=====
Train      Real: 28,000 ( 54.7%) Fake: 23,199 ( 45.3%) Total: 51,199
Validation Real:  3,500 ( 54.7%) Fake:  2,900 ( 45.3%) Total:  6,400
Test       Real:  3,500 ( 54.7%) Fake:  2,901 ( 45.3%) Total:  6,401

Per-Domain Breakdown
-----

Train:
label      0      1
domain
face_main  12000   7200
nano_banana 16000  15999
```

```
Validation:
label          0      1
domain
face_main     1500    900
nano_banana   2000    2000
```

```
Test:
label          0      1
domain
face_main     1500    901
nano_banana   2000    2000
```

[OK] Train/Test class ratio gap is small (0.01%).

```
=====
Split Leakage Check (duplicate filepaths across splits)
=====
```

[OK] No filepath overlap between train/val/test splits.

	filepath	label	domain
0	/kaggle/input/datasets/ahnuf05/ai-imagewith-na...	0	nano_banana
1	/kaggle/input/datasets/mohannadaymansalah/stab...	1	face_main
2	/kaggle/input/datasets/ahnuf05/ai-imagewith-na...	1	nano_banana
3	/kaggle/input/datasets/mohannadaymansalah/stab...	1	face_main
4	/kaggle/input/datasets/ahnuf05/ai-imagewith-na...	1	nano_banana

+ Code

+ Markdown

3. Face Detection & Alignment Preprocessing (*Vishaka*

```
=====
Face Preprocessing (RetinaFace, one-time, cached)
=====
```

Preprocessing train: 100%  51199/51199 [00:03<00:00, 14320.37it/s]

train: 51199 processed, 0 fallbacks.

[OK] No cache-key collisions or cross-split duplicates detected.

Preprocessing val: 100%  6400/6400 [00:00<00:00, 13603.46it/s]

val: 6400 processed, 0 fallbacks.

[OK] No cache-key collisions or cross-split duplicates detected.

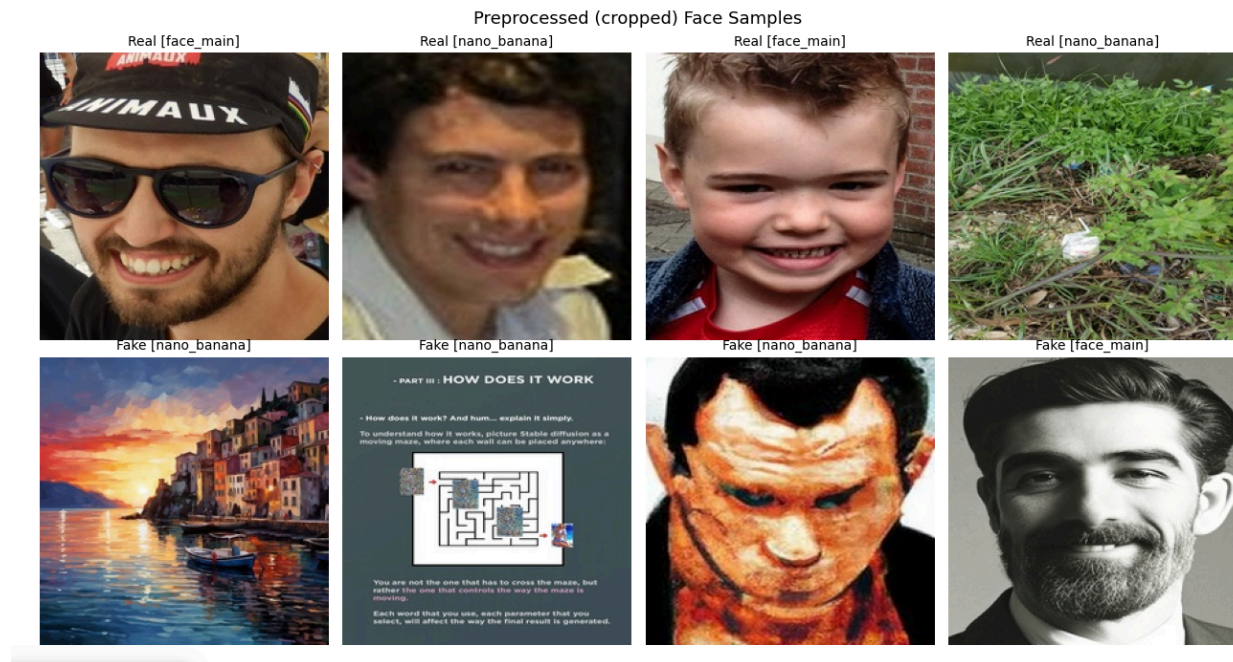
Preprocessing test: 100%  6401/6401 [11:10<00:00, 7.71it/s]

test: 6401 processed, 0 fallbacks.

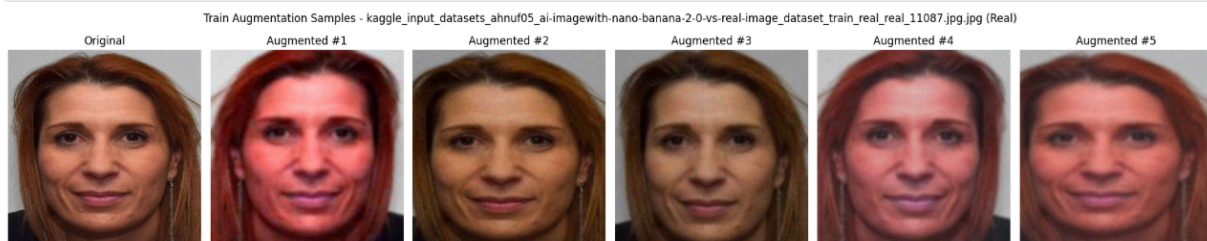
[OK] No cache-key collisions or cross-split duplicates detected.

Preprocessing complete. Training will now load pre-cropped faces.

Preprocessing complete. Training will now load pre-cropped faces.



4. Robust Transformation Pipeline (*Vishaka*)



5. Dataset & DataLoaders

```
=====
DataLoader Ready
=====

Train Images : 51,199
Val Images   : 6,400
Test Images  : 6,401

Batch Shape : torch.Size([64, 3, 224, 224])
Labels Shape: torch.Size([64])
Tensor Type : torch.float32

+ Code    + Markdown
```

6. Model — MobileNetV3-Large

```
=====
Model Ready
=====
Total Parameters      : 4,204,594
Trainable Parameters  : 1,232,642
+ Code + Markdown
```

8. Stage 1 & Stage 2 Training (*Raunak, using Rohit's tuned hparams*)

```
=====
Stage 1 : Training Classifier
=====
Epoch [1/3] | Train Loss 0.3250 | Train Acc 85.98% | Val Loss 0.6487 | Val Acc 68.73% | Val AUC 0.8375 | Time 360.2s
Best model saved (68.73%)
Epoch [2/3] | Train Loss 0.2753 | Train Acc 88.56% | Val Loss 0.6140 | Val Acc 70.30% | Val AUC 0.8342 | Time 355.2s
Best model saved (70.30%)
Epoch [3/3] | Train Loss 0.2500 | Train Acc 89.68% | Val Loss 0.6399 | Val Acc 69.91% | Val AUC 0.8271 | Time 354.1s

Stage 1 : Training Classifier Complete - Best Val Acc: 70.30%

Total Feature Blocks : 17
Unfreezing Blocks    : 12 -> 16
Trainable Parameters : 3,798,226
=====
Stage 2 : Fine-Tuning
=====
Epoch [1/7] | Train Loss 0.2202 | Train Acc 91.14% | Val Loss 0.7216 | Val Acc 68.34% | Val AUC 0.8474 | Time 384.7s
Best model saved (68.34%)
Epoch [2/7] | Train Loss 0.1827 | Train Acc 92.77% | Val Loss 0.7833 | Val Acc 67.47% | Val AUC 0.8516 | Time 357.8s
Epoch [3/7] | Train Loss 0.1613 | Train Acc 93.62% | Val Loss 0.8588 | Val Acc 66.89% | Val AUC 0.8651 | Time 351.8s
Epoch [4/7] | Train Loss 0.1500 | Train Acc 94.06% | Val Loss 0.8214 | Val Acc 67.95% | Val AUC 0.8688 | Time 357.1s
Epoch [5/7] | Train Loss 0.1389 | Train Acc 94.67% | Val Loss 0.8517 | Val Acc 67.66% | Val AUC 0.8710 | Time 344.9s
Epoch [6/7] | Train Loss 0.1365 | Train Acc 94.71% | Val Loss 0.8609 | Val Acc 67.22% | Val AUC 0.8710 | Time 350.0s
Epoch [7/7] | Train Loss 0.1339 | Train Acc 94.77% | Val Loss 0.8375 | Val Acc 67.89% | Val AUC 0.8737 | Time 333.3s

Stage 2 : Fine-Tuning Complete - Best Val Acc: 68.34%

=====
Training Finished
=====
Best Validation Accuracy : 68.34%
+ Code + Markdown
```

9. Final Evaluation

9. Final Evaluation

100%  101/101 [00:13<00:00, 1.77it/s]

=====
Test Results
=====

Test Loss : 0.7416
Test Accuracy : 66.21%
Test F1 : 0.7079
Test AUC : 0.8414

Classification Report

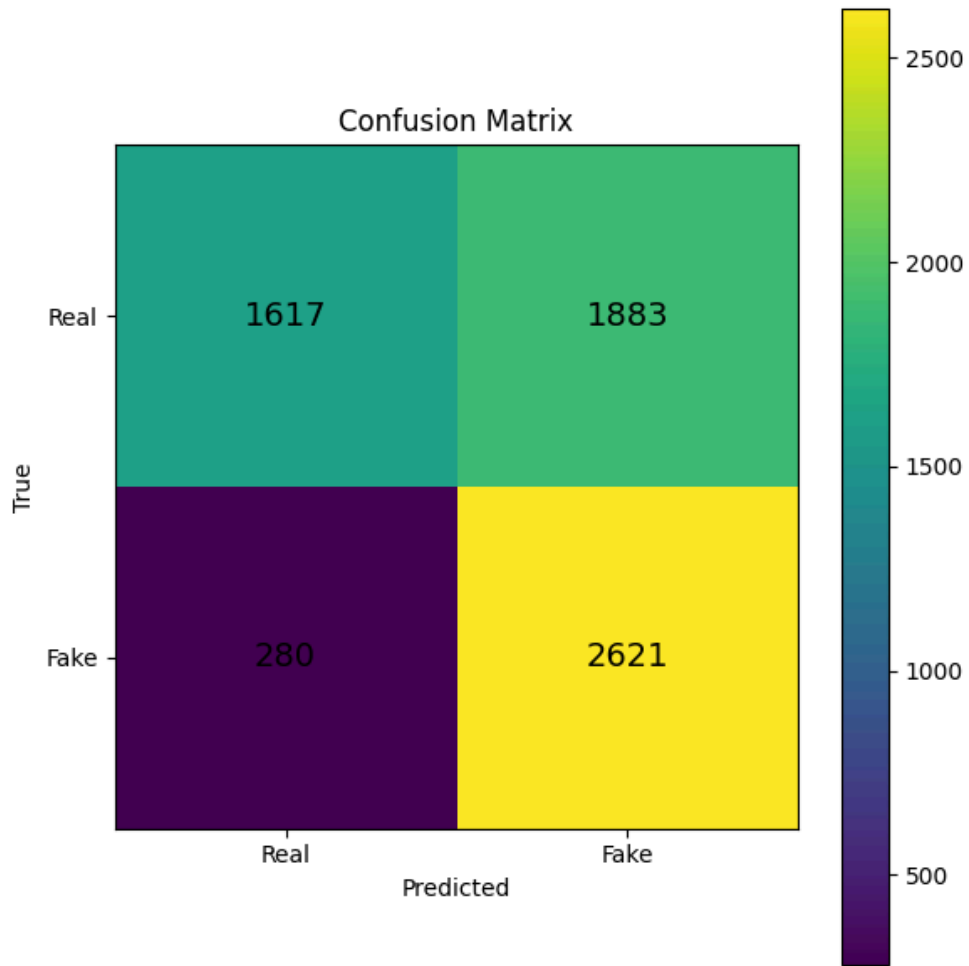
	precision	recall	f1-score	support
Real	0.8524	0.4620	0.5992	3500
Fake	0.5819	0.9035	0.7079	2901
accuracy			0.6621	6401
macro avg	0.7172	0.6827	0.6536	6401
weighted avg	0.7298	0.6621	0.6485	6401

Confusion Matrix

[[1617 1883]
[280 2621]]

 2500

```
[[1617 1883]  
 [ 280 2621]]
```

[+ Code](#)[+ Markdown](#)

9b. Augmentation Ablation Training - No-Augmentation Model (*Vishakha*)

```
#####
# No-Augmentation Model (ROBUST_TRAIN_AUG=False)
#####
DataLoader Ready
=====
Train Images : 51,199
Val Images   : 6,400
Test Images  : 6,401
=====
No-Augmentation Model - Stage 1
=====
Epoch [1/3] | Train Loss 0.2110 | Train Acc 91.58% | Val Loss 0.1603 | Val Acc 93.62% | Val AUC 0.9840 | Time 60.6s
Best model saved (93.62%)
Epoch [2/3] | Train Loss 0.1655 | Train Acc 93.47% | Val Loss 0.1440 | Val Acc 94.27% | Val AUC 0.9872 | Time 62.5s
Best model saved (94.27%)
Epoch [3/3] | Train Loss 0.1456 | Train Acc 94.36% | Val Loss 0.1386 | Val Acc 94.70% | Val AUC 0.9880 | Time 59.9s
Best model saved (94.70%)

No-Augmentation Model - Stage 1 Complete - Best Val Acc: 94.70%
=====
No-Augmentation Model - Stage 2
=====
Epoch [1/7] | Train Loss 0.1190 | Train Acc 95.44% | Val Loss 0.0979 | Val Acc 96.42% | Val AUC 0.9936 | Time 63.1s
Best model saved (96.42%)
Epoch [2/7] | Train Loss 0.0850 | Train Acc 96.80% | Val Loss 0.0843 | Val Acc 97.12% | Val AUC 0.9952 | Time 63.3s
Best model saved (97.12%)
Epoch [3/7] | Train Loss 0.0667 | Train Acc 97.60% | Val Loss 0.0764 | Val Acc 97.16% | Val AUC 0.9961 | Time 62.8s
Best model saved (97.16%)
Epoch [4/7] | Train Loss 0.0557 | Train Acc 98.08% | Val Loss 0.0707 | Val Acc 97.47% | Val AUC 0.9967 | Time 62.9s
Best model saved (97.47%)
Epoch [5/7] | Train Loss 0.0489 | Train Acc 98.26% | Val Loss 0.0690 | Val Acc 97.53% | Val AUC 0.9968 | Time 62.5s
Best model saved (97.53%)
Epoch [6/7] | Train Loss 0.0444 | Train Acc 98.50% | Val Loss 0.0678 | Val Acc 97.61% | Val AUC 0.9969 | Time 61.4s
Best model saved (97.61%)
Epoch [7/7] | Train Loss 0.0434 | Train Acc 98.51% | Val Loss 0.0672 | Val Acc 97.58% | Val AUC 0.9970 | Time 61.7s

No-Augmentation Model - Stage 2 Complete - Best Val Acc: 97.61%

No-Augmentation Model Complete - Best Val Acc: 97.61%

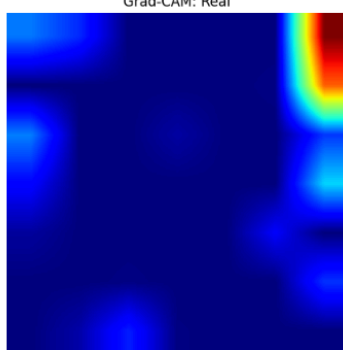
=====
Augmentation Ablation - Test Set Comparison (Priority 1)
=====
      Model  Accuracy (%)   F1   AUC
With Augmentation      66.21 0.7079 0.8414
Without Augmentation   97.56 0.9729 0.9970
```

[+ Code](#) [+ Markdown](#)

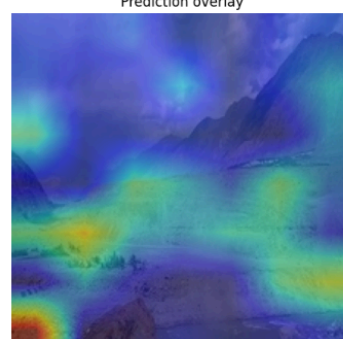
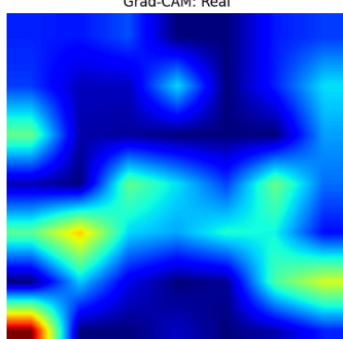
10. Grad-CAM Explainability (*Somendu*

Grad-CAM figures saved to /kaggle/working/outputs

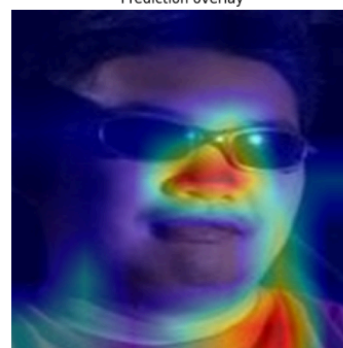
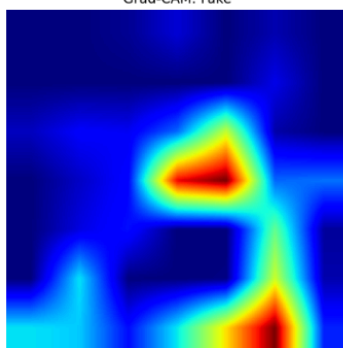
[CORRECT] kaggle_input_datasets_ahnuf05_ai-imagewith-nano-banana-2-0-vs-real-image_dataset_test_real_real_01936.jpg.jpg | Prediction: Real | True: Real
Real: 56.79% | Fake: 43.21%
Grad-CAM: Real



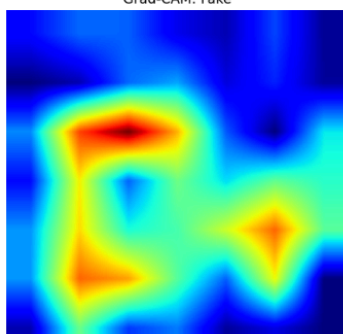
[CORRECT] kaggle_input_datasets_ahnuf05_ai-imagewith-nano-banana-2-0-vs-real-image_dataset_test_real_real_00531.jpg.jpg | Prediction: Real | True: Real
Real: 98.12% | Fake: 1.88%
Grad-CAM: Real



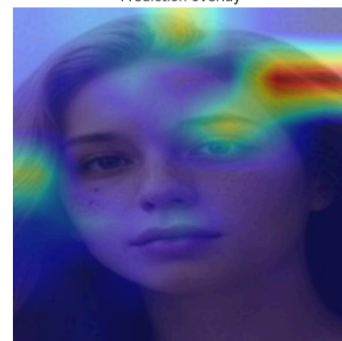
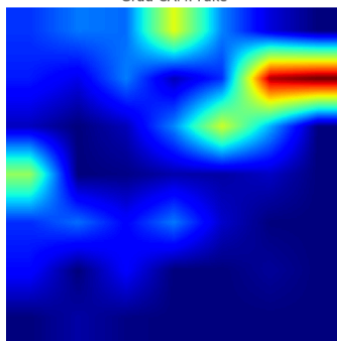
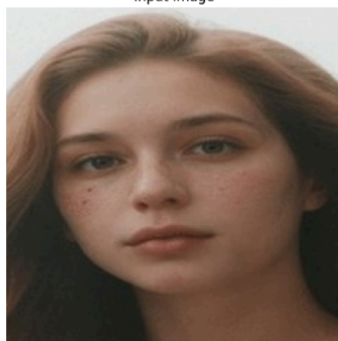
[INCORRECT] kaggle_input_datasets_ahnuf05_ai-imagewith-nano-banana-2-0-vs-real-image_dataset_test_real_real_00203.jpg.jpg | Prediction: Fake | True: Real
Real: 27.79% | Fake: 72.21%
Grad-CAM: Fake



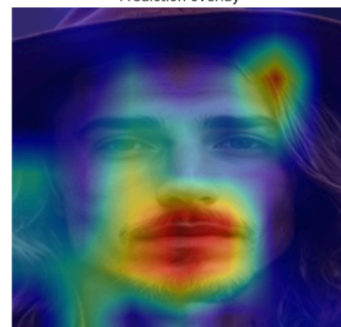
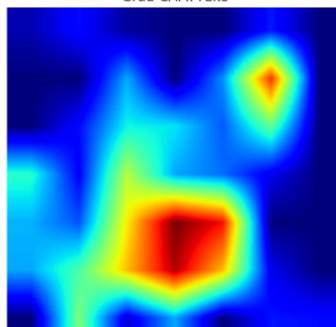
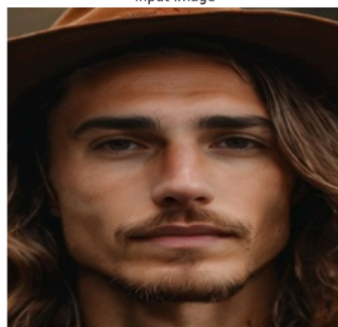
[INCORRECT] kaggle_input_datasets_ahnuf05_ai-imagewith-nano-banana-2-0-vs-real-image_dataset_test_real_real_01694.jpg.jpg | Prediction: Fake | True: Real
Real: 2.37% | Fake: 97.63%
Grad-CAM: Fake



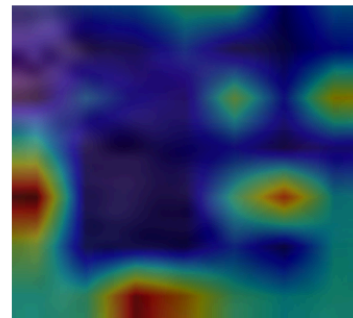
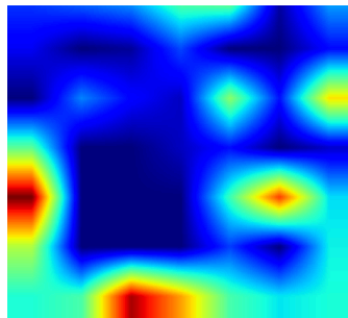
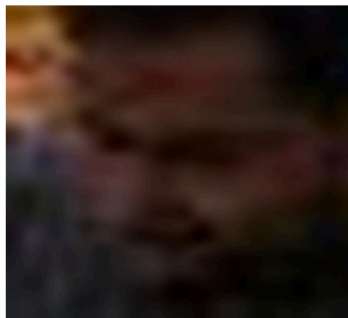
[CORRECT] kaggle_input_datasets_ahnuf05_ai-imagewith-nano-banana-2-0-vs-real-image_dataset_test_fake_fake_00151.jpg.jpg | Prediction: Fake | True: Fake
Real: 14.97% | Fake: 85.03%
Grad-CAM: Fake



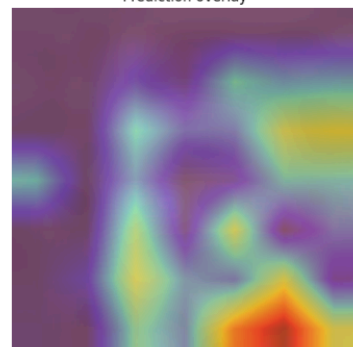
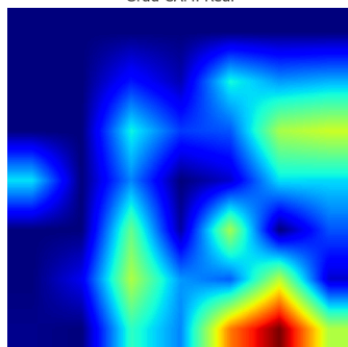
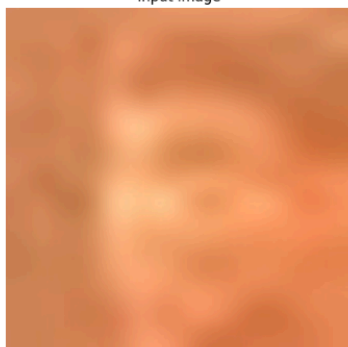
[CORRECT] kaggle_input_datasets_mohannadaymansalah_stable-diffusion-dataaaaaaaaaa_stable-diffusion-face-dataset_1024_man_men_1426.jpg.jpg | Prediction: Fake | True: Fake
Real: 0.51% | Fake: 99.49%
Grad-CAM: Fake



四



[INCORRECT] kaggle_input_datasets_ahnuf05_ai-imagewith-nano-banana-2-0-vs-real-image_dataset_test_fake_fake_00429.jpg.jpg | Prediction: Real | True: Fake
Real: 69.03% | Fake: 30.97%
Grad-CAM: Real



Grad-CAM figures saved to /kaggle/working/outputs

[+ Code](#) [+ Markdown](#)

11. Robustness Evaluation (*Vishaka*)

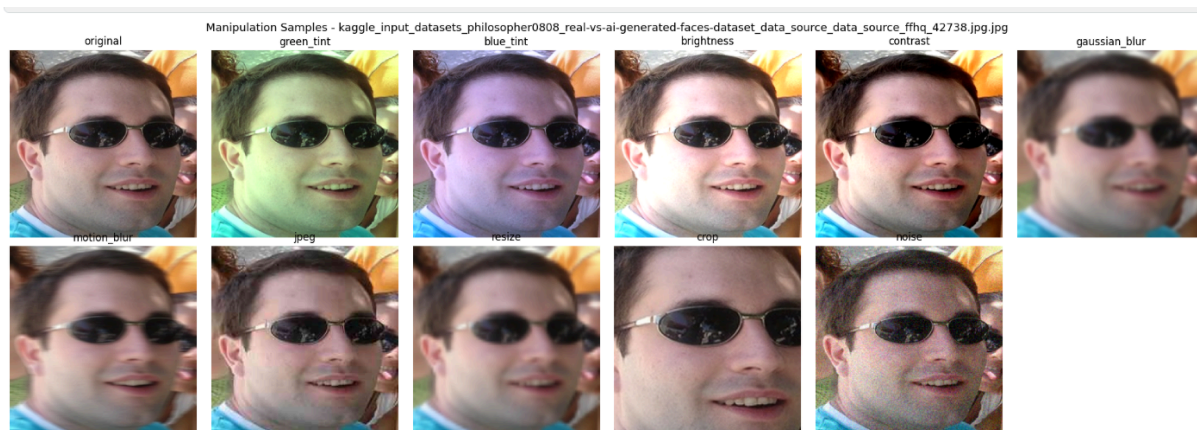
```

=====
Robustness Evaluation (corruption accuracy)
=====

Loading widget...
Loading widget...
Loading widget...
Loading widget...
Loading widget...
Manipulation Accuracy
  Original  74.00%
    Tint    72.33%
    Blur    62.00%
    Noise   83.00%
    Jpeg     64.00%

```

12b. Image Manipulation Testing (*Aman*)



```

Loading widget...
Manipulation  Accuracy (%)  Delta vs Original (pp)
  original          74.00           0.00
  green_tint        72.33          -1.67
  blue_tint         76.00           2.00
  brightness        60.00        -14.00
  contrast          55.67        -18.33
  gaussian_blur     62.00        -12.00
  motion_blur       64.67          -9.33
  jpeg              64.00        -10.00
  resize            62.33        -11.67
  crop              67.67          -6.33
  noise             85.00         11.00

```

12c. Cross-Domain Testing (*Raunak*)

Real

Real

Cross-Domain Samples (nano_banana)

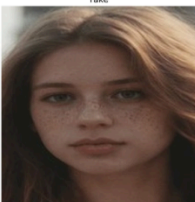
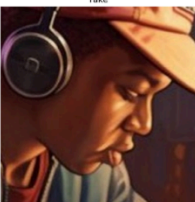
Fake

Fake

Fake





=====

Cross-Domain Testing (Priority 4 - Raunak)

=====

Domain	N	Accuracy (%)	F1	AUC
face_main	2401	59.89	0.6502	0.8993
nano_banana	4000	70.00	0.7420	0.8379

Operational boundary: accuracy changes by ~10.11 percentage points moving from the core face_main domain to cross-domain (nano_banana) data.
Interpretation: use this gap as the confidence discount to apply when the model is run on images outside the core face real-vs-AI domain.

[+ Code](#)
[+ Markdown](#)

12. Hyperparameter Sweep (Rohit)

```
=====
Hyperparameter Sweep (fresh run on current dataset mix)
=====
```

```
=====
Experiment : E0_Baseline
=====
```

```
Loading image paths...
Found 70,000 Real images
Found 9,001 Fake images
```

```
Dataset Summary
```

```
label
0      15000
1       9001
```

```
Total Images : 24,001
```

```
Extra Dataset detected, merging...
```

```
=====
Final Dataset
=====
```

```
Train      : 51,199
Validation : 6,400
Test       : 6,401
```

```
=====
Stratification / Class-Balance Check
=====
```

```
Train      Real: 28,000 ( 54.7%)  Fake: 23,199 ( 45.3%)  Total: 51,199
Validation Real:  3,500 ( 54.7%)  Fake:  2,900 ( 45.3%)  Total:  6,400
Test       Real:  3,500 ( 54.7%)  Fake:  2,901 ( 45.3%)  Total:  6,401
```

```
Per-Domain Breakdown
-----
```

Per-Domain Breakdown

Train:		
label	0	1
domain		
face_main	12000	7200
nano_banana	16000	15999

Validation:		
label	0	1
domain		
face_main	1500	900
nano_banana	2000	2000

Test:		
label	0	1
domain		
face_main	1500	901
nano_banana	2000	2000

50% Train / Test split ratio

```
=====
Epoch [1/3] | Train Loss 0.2331 | Train Acc 90.41% | Val Loss 0.7720 | Val Acc 66.81% | Val AUC 0.8459 | Time 330.8s
Best model saved (66.81%)
Epoch [2/3] | Train Loss 0.1907 | Train Acc 92.45% | Val Loss 0.8183 | Val Acc 66.39% | Val AUC 0.8533 | Time 320.9s
Epoch [3/3] | Train Loss 0.1730 | Train Acc 93.08% | Val Loss 0.8613 | Val Acc 65.88% | Val AUC 0.8565 | Time 321.4s
```

Stage 2 (exp E0_Baseline) Complete - Best Val Acc: 66.81%

100%  101/101 [00:07<00:00, 21.60it/s]

```
=====
Test Results
=====
```

```
Test Loss      : 0.7842
Test Accuracy   : 65.22%
Test F1         : 0.7051
Test AUC        : 0.8424
```

Classification Report

	precision	recall	f1-score	support
Real	0.8632	0.4326	0.5763	3500
Fake	0.5726	0.9173	0.7051	2901
accuracy			0.6522	6401
macro avg	0.7179	0.6749	0.6407	6401
weighted avg	0.7315	0.6522	0.6347	6401

Confusion Matrix
[[1514 1986]
[240 2661]]

```
=====
Experiment : E1a_LR_1e4
=====
```

Loading image paths...

[+ Code](#)

[+ Markdown](#)